

Pendulum Lab V10 Modernization

Typed, tested, browser-based chaotic pendulum simulation with legacy compatibility and a modern TypeScript physics path.

- TypeScript
- Vite
- Numerical Methods
- Playwright
- Scientific Validation

WHAT CHANGED

- Direct file execution now falls back to main-thread physics so the pendulum keeps moving.
- TypeScript physics core powers the served default double-pendulum path.
- Canonical theta/p Hamiltonian utilities and residual-reporting midpoint diagnostics were added.
- Strict JSON import validation and legacy risk audits reduce browser security risk.

WHY IT MATTERS

Chaotic systems expose hidden numerical and reproducibility problems quickly. The same engineering discipline applies to semiconductor and device-physics workflows where solver assumptions, drift, and validation gaps can change interpretation.

VERIFICATION

Automated checks cover typecheck, unit tests, motion E2E, modern demo E2E, build, benchmark report generation, validation report generation, and weighted legacy-risk reduction.

KNOWN LIMITS

Symplectic claims require canonical coordinates, $\gamma = 0$, and converged residuals. Lyapunov and bifurcation views are finite-time diagnostics and require disclosed parameters.